

Information Theory and Cognition

Problem Set 3

Due: Friday 2/20 (before midnight)

1 The information value of a question

Suppose you and a friend are playing the following game: your friend draws a number $X \in \{1, \dots, n\}$ from a known distribution $p(x)$. Your goal is to guess the number. You are allowed to ask questions of the form “is $X \in S$?” for a subset $S \subseteq \{1, \dots, n\}$, and let Y be the answer to the question, i.e.,

$$Y = \begin{cases} 1 & X \in S \\ 0 & X \notin S \end{cases}$$

Suppose that $p(X \in S) = \alpha$.

- 1.1 Find the decrease in uncertainty, i.e., $H(X) - H(X|Y)$.
- 1.2 Assume now that $p(x)$ is uniform. Give an example of a question that maximizes the decrease in uncertainty.

2 Data Processing Inequality

- 2.1 Let $X \rightarrow Y \rightarrow Z$ form a Markov chain. Show that $Z \rightarrow Y \rightarrow X$ is also a Markov chain. Conclude that $I(X; Z) \leq \min\{I(X; Y), I(Y; Z)\}$.
- 2.2 We will now show a version of the data processing inequality for the KL divergence. Let $p(x)$ and $q(x)$ be two distributions over $\mathcal{X}$. Let $Y \sim r(y|x)$ be a processed version of X , where $r(y|x)$ is a conditional distribution characterizing the processing channel. Let $p(x, y) = p(x)r(y|x)$ and $q(x, y) = q(x)r(y|x)$. Show that

$$D[p(x) \| q(x)] \geq D[p(y) \| q(y)].$$

Intuitively, this means that processing data generated by two distributions makes the distributions less distinguishable.

3 KL divergence

Find the KL divergence between the following distributions:

- 3.1 $p(x) = U(\{1, \dots, m\})$, $q(x) = U(\{1, \dots, n\})$, where U is a uniform distribution. Discuss what happens when $m < n$, and when $m > n$.

- 3.2 $f(x) = \mathcal{N}(x; \mu_1, \sigma_1^2)$, $g(x) = \mathcal{N}(x; \mu_2, \sigma_2^2)$. Discuss what happens when $\sigma_1 = \sigma_2$, and when $\mu_1 = \mu_2$.
- 3.3 $p(x) = \frac{1}{z_1} e^{\beta_1 U(x)}$, $q(x) = \frac{1}{z_2} e^{\beta_2 U(x)}$, where $U(x)$ is a utility function over $X \in \{1, \dots, n\}$, $\beta_i \geq 0$, and $z_i = \sum_x e^{\beta_i U(x)}$ are the normalization factors. You can express the KL divergence in terms of $u = \mathbb{E}[U(X)]$ as long as you explain which distribution is used for taking the expectation. Discuss what happens when (a) $\beta_2 = 0$, (b) $\beta_2 \rightarrow \infty$ while β_1 is fixed, and (c) $\beta_1 \rightarrow \infty$ while β_2 is fixed.